

NEET Important Chapter – Electromagnetic Induction

Electromagnetic Induction – An Important Concept for NEET

Search your topic here...

From previous chapters, we know that current produces magnetic fields. Is the reverse of this true? Can a magnetic field produce electric current? The answer is yes. For a very long time, electricity and magnetism were considered a separate field but it is now known that they are interrelated.

It is found that currents are produced in closed coils when placed in a varying magnetic field. The phenomenon in which current is induced by a changing magnetic field is known as Electromagnetic Induction. The current thus produced is called the induced current. If the current is produced, it must be due to some emf produced in the circuit, which is known as induced emf.

In this chapter, students will learn about the central principle of electromagnetic induction i.e. Faraday's laws. Students will also learn about Lenz's law, which helps to predict the direction of induced current.

Going further in this chapter, students will also learn about the construction and working of devices based on electromagnetic induction such as power generators, dynamos, transformers, etc.

Now, let us move on to the important concepts and formulae related to the NEET exam along with a few solved examples.

Important Topics of Electromagnetic Induction

- Electromagnetic Induction
- Magnetic Flux
- Faraday's Law
- Lenz's Law
- Motional Electromotive Force
- Self-inductance
- Mutual Inductance
- Growth and Decay of Current in an L-R Circuit
- Oscillations in L-C Circuit

Important Concepts of Electromagnetic Induction

Name of the Concept	Key Points of the Concept
1. Magnetic Flux	• Magnetic Flux through any surface held in a magnetic field is measured as the total number of magnetic field lines crossing the surface. • It is a scalar quantity. • Mathematically, it is given by • SI unit of magnetic flux is Weber (Wb) • Dimensional formula is

2. Faraday's Law of Magnetic Induction	• Faraday's Law of Magnetic Induction states that the induced emf in a closed loop is equal to the negative of the time rate of change of magnetic flux through the loop. Mathematically, $\epsilon = - \frac{d\phi}{dt}$ - Or, $\epsilon = - N \frac{d\phi}{dt}$
3. Lenz's law	• This law states that the direction of magnetic induction is such to oppose the cause of the effect.
4. Motional Emf	• Let a conducting rod of length (l) be moving with the uniform velocity (v) perpendicular to the magnetic field (B). Then the magnitude of induced emf will be • If the rod is moving such that it makes an angle θ with the direction of magnetic field, then
5. Self- Induction	• Self- induction is the phenomenon due to which an induced emf is set up in a coil or a circuit whenever the current passing through it changes.

6. Inductance	- Inductance is the inherent property of electrical circuits. - An inductor is said to be an ideal inductor if its resistance is zero. - An inductor does not change current but opposes the change in the current. - Emf induced due to self inductance is given by: — - The SI unit of inductance is Henry.
7. Potential energy of an inductor	In building a steady current in an electrical circuit, the emf has to do some work, which is given by —
8. Mutual Inductance	Mutual inductance is the phenomenon due to which an emf is induced in the coil when the current flowing through a neighbouring coil changes.
9. Eddy Currents	Currents induced in the body of the bulk of the conductors due to change in the magnetic flux linked to them are called eddy currents.
10. Oscillations in L-C Circuit	If a charged capacitor C is short-circuited through an inductor L, the charge and current in the circuit start oscillating simple harmonically.

List of Important Formulae

Sl. No	Name of the Concept	Formula
1.	Magnetic flux	Where, B = Magnetic field, A = Area and θ = angle between magnetic field and area.
2.	Faraday's law of magnetic induction	$\text{Or, } \frac{d\Phi}{dt}$ Where, N= number of turns, B = Magnetic field, A = Area, θ = angle between magnetic field and area and t = time.
3.	Motional emf in loop	Where, B = Magnetic field, l = length of the conductor and v = velocity of the conductor.
4.	Induced current in loop due to motional emf	$\frac{B l v}{R}$ Where, B = Magnetic field, l = length of the conductor, v = velocity of the conductor and R = Resistance
5.	Magnetic force on loop due to motional emf	$B I l$ Where, B = Magnetic field, l = length of the conductor and I = Current.

6.	Rotational emf	— Where, B = Magnetic field, l = length of the conductor and ω = angular velocity of the conductor.
7.	Emf due to self-induction	— Where, L = self- inductance and I = current
8.	Potential energy of an Inductor	— Where, L = self- inductance and I = current
9.	Emf due to mutual-induction	— Where, M = mutual- inductance and I = current
10.	Inductors in series placed far apart	
11.	Inductors in series placed close to each other	
12.	Inductors in parallel	— — —
13.	Angular frequency of L-C Oscillations	— Where, ω = Angular frequency, L = Inductance and C = Capacitance

Solved Examples

1. The current in a long current carrying wire is $I = 2t$. A conducting loop is placed to the right of this wire. Find magnetic flux passing through the loop.

Sol:

In this situation, there is no motion of the conductor and the magnetic field changes with the distance from the current carrying conductor. So, the magnetic flux will be obtained by using an integration method.

Consider an element of thickness, dx at a distance x from the wire as shown in the figure.

At distance, x magnetic field (B) is

— —

Now, let us find the area of the strip

Therefore, a small magnetic flux passing through the loop will be

— —

The total magnetic flux is

Also, it is given that $I = 2t$

Hence, the magnetic flux passing through the loop is

Key point: In such cases where there is no motion and magnetic field is varying as a function of distance, the integration method comes in handy.

2. A square loop of side l having resistance R moves with constant velocity v , through a constant magnetic field as shown in figure. Find the magnitude and direction of induced current in the situation when

- 1. The loop enters into the magnetic field.**
- 2. It is moving in a magnetic field.**
- 3. It is coming out of the magnetic field.**

Sol:

1. Here the side ad is outside the field. Therefore, no magnetic flux will be induced in side ad. Now, sides ab and cd are parallel to the velocity of the loop and no current is induced when the velocity is parallel to the velocity of the loop. So, the emf will only be induced in the side cb as shown in figure.

Hence, net emf induced will be

The current in the loop $\rightarrow$ = $\rightarrow$. The current will be counterclockwise

2. When the loop is completely submerged in the field, there is no change in the flux w.r.t. time. So, there will not be an induction of current.

3. When the loop starts coming out of the magnetic field. The emf will be induced in side ad only. Using Lenz's law, the direction of current will be clockwise.

Hence, the net emf induced will be

The current in the loop $\rightarrow$ = $\rightarrow$.

Key Point: Always use Lenz's law to find the direction of induced current.

Previous Year Questions from NEET Paper

1. The magnetic flux linked with a coil (in Wb) is given by the equation $\phi = 10t^2 - 4t$. The magnitude of induced emf in the coil at the fourth second will be (NEET 2020)

- a. 33 V
- b. 43 V
- c. 108 V
- d. 10 V

Sol:

The magnetic flux linked with the coil is given as

The magnitude of the induced emf is

$$\frac{d\phi}{dt} = \frac{d(10t^2 - 4t)}{dt}$$

$$\text{At } t = 3s, \quad \epsilon = 33V,$$

$$\text{At } t = 4s, \quad \epsilon = 43V,$$

Therefore, the induced emf will be:

$$\epsilon = 10 V$$

Hence, the correct answer is option (d)

Trick: The rate of change of magnetic flux is emf induced in the conductor.

2. A wheel with 20 metallic spokes, each 1 m long, is rotated with the speed of 120 rpm in a plane perpendicular to a magnetic field of 0.4 G. The induced emf between axle and rim of the wheel will be (1G = 10^{-4} T) (NEET 2020)

- a.
- b.

--

c.

d.

Sol:

Given,

Magnetic field (B) = 0.4 G = _____ ,

Frequency (f) = 120 rpm = _____ rps = 2 Hz,

The induced emf between the axle and the rim of the wheel is given as

—

— =

Option (a) is the correct answer.

Trick: Rotational emf is given by —

Practice Questions

1. A coil formed by wrapping 50 turns of a wire in the shape of a square is positioned in a magnetic field so that the normal to the plane of the coil makes an angle of 30° , with the direction of the field. When the magnetic field is increased uniformly from $200 \mu\text{T}$ to $600 \mu\text{T}$ in 0.4 s, an emf of magnitude 80.0 mV is induced in the coil. What is the total length of the wire? **(Ans. 272 m)**

2. A square frame with side a and a long straight wire carrying a current i are located in the same plane as shown in the figure. The frame translates to the right with a constant velocity v. Find the emf induced in the frame as a function of distance x. **(Ans. —————)**

Conclusion

In this article, we have provided important information regarding the chapter Electromagnetic Induction such as important concepts, formulae, etc. Students should work on more solved examples for securing good grades in the [NEET 2022](#) exam.

Hi Aditya,

Best courses for you

**Full syllabus LIVE
courses**

Starting from ₹32,400

**One-to-one LIVE
classes**

Starting from ₹0/hr

Book your **Free Demo** session

Get a flavour of LIVE classes here at Vedantu

[Book a free demo](#)

Vedantu Improvement Promise

We promise improvement in marks or get your fees back. T&C Apply*